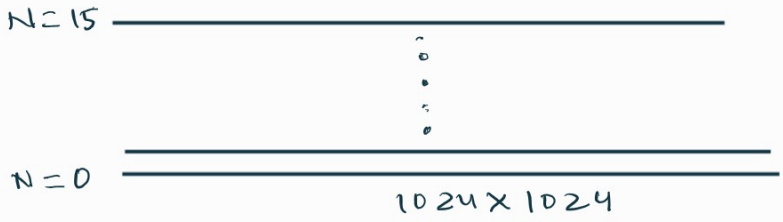
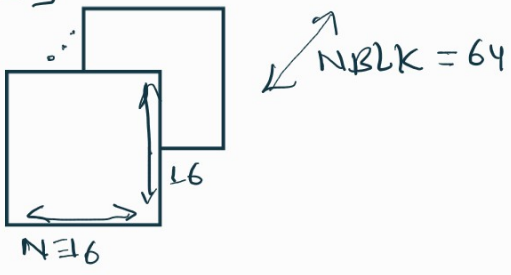
▷ input format

- int (*data)[matrix_size × matrix_size]
  1024 1024

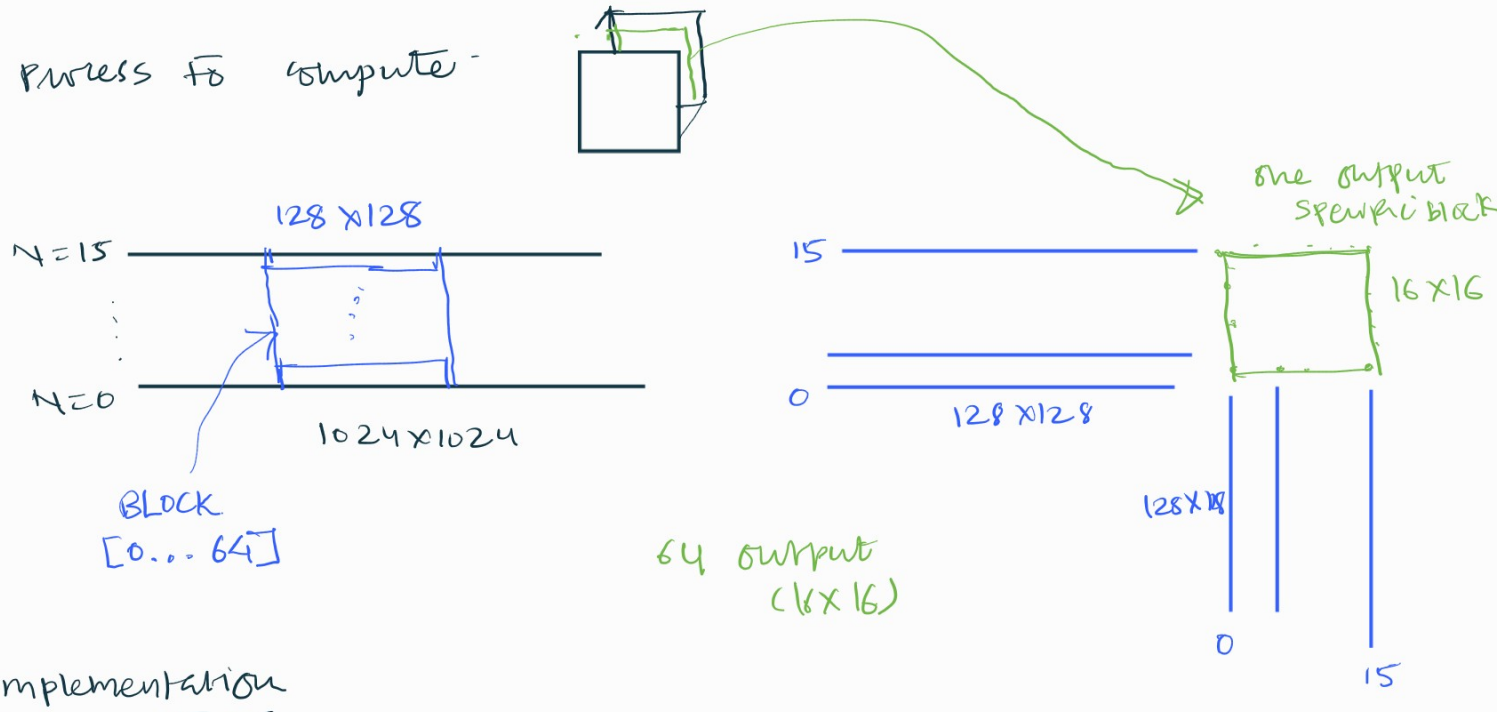


▷ output format

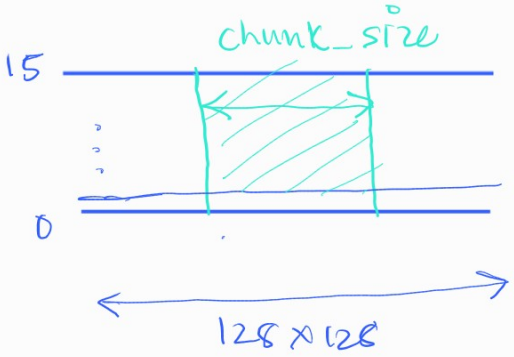
int (*output)[N][N]



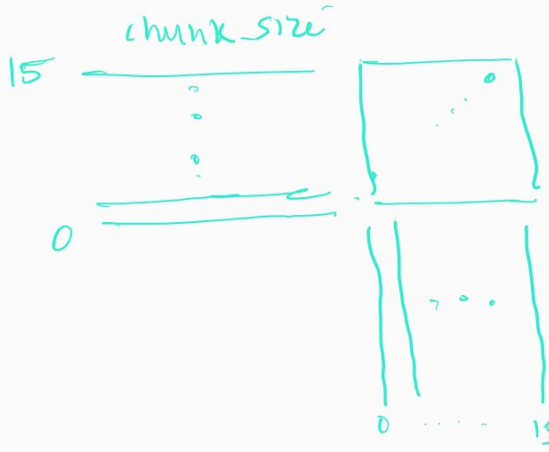
▷ process to compute:



▷ implementation



▷ Fit [16][chunk_size] to shared memory per block.

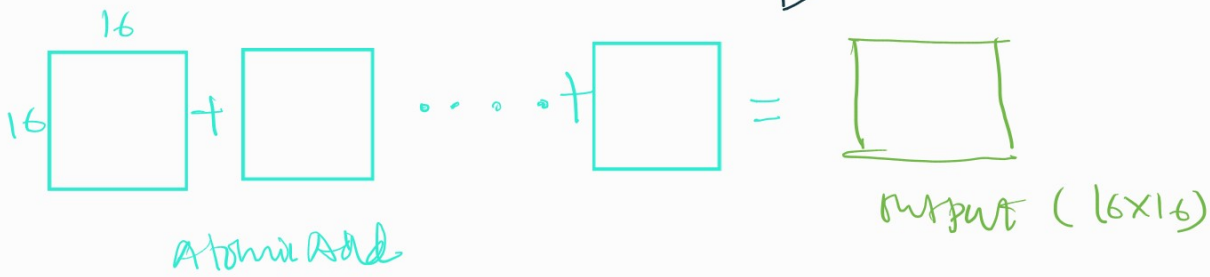


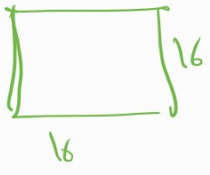
▷ calculate transpose

[16][chunksize]

x [chunksize][16]

which is partial.



▷ Load  to

Global memmory

▷ Grid dimension

$$NBLK \times \underbrace{\left(\frac{BLOCK_SIZE}{128} * \frac{BLOCK_SIZE}{128}\right) / CHUNK_SIZE}_{CHUNKS}$$

(64)

dim3 Grid (NBLK, CHUNKS)

dim3 block ($\frac{N}{16}$, $\frac{N}{16}$)

Each block  responsibility is to calculate

transpose and accumulate to sum.

No of threads : 16x16 = 256

each threads calculate element at position (i, j)



- coalescing ✓
- Bank conflict ✓

- TODO, Hermitian matrix, symmetric